

Assignment - 4

Question-1

Greedy Search

```
↔ ('S', 'A')
   ('S', 'D')
   ('A', 'B')
   ('A', 'C')
   ('B', 'C')
   ('B', 'E')
   ('C', 'G')
   ('D', 'B')
   ('D', 'E')
   ('E', 'G')
Adj matrix = [[0. 1. 0. 0. 1. 0. 0.]
 [0. 0. 1. 1. 0. 0. 0.]
 [0. 0. 0. 1. 0. 1. 0.]
 [0. 0. 0. 0. 0. 0. 1.]
 [0. 0. 1. 0. 0. 1. 0.]
 [0. 0. 0. 0. 0. 0. 1.]
 [0. 0. 0. 0. 0. 0. 0.]]
0
[-1, 0, -1, -1, 0, -1, -1]
4
[-1, 0, 4, -1, 0, 4, -1]
5
[-1, 0, 4, -1, 0, 4, 5]
6
Goal node6 found.
Path from S to G:
['S', 'D', 'E', 'G']
```

Question- 2

A* Tree Search

```
↔ Found the goal node.
S -> D -> B -> E -> G
Cost obtained finally is this = 7.0
```

Question- 3

A* Graph Search

```
Adjacency Matrix
[[ 0.  3.  0.  0.  2.  0.  0.]
 [ 0.  0.  5. 10.  0.  0.  0.]
 [ 0.  0.  0.  2.  0.  1.  0.]
 [ 0.  0.  0.  0.  0.  0.  4.]
 [ 0.  0.  1.  0.  0.  4.  0.]
 [ 0.  0.  0.  0.  0.  0.  3.]
 [ 0.  0.  0.  0.  0.  0.  0.]]
Found the goal node.
S
D
B
E
G
Cost obtained finally is this = 7.0
```

Question - 4

```
Enter size: 8
Enter start row: 1
Enter start column: 1
** Knight's Tour **
Starting position: (1, 1)
 1 26 15 24 29 36 13 32
16 23 28 35 14 31 40 37
27  2 25 30 61 38 33 12
22 17 62 45 34 41 50 39
 3 46 21 60 49 64 11 42
18 57 48 63 44 53  8 51
47  4 55 20 59  6 43 10
56 19 58  5 54  9 52  7
```

Question - 5

Small-Input

```
testcase Total = 10
submarine.out: [-1, -1, 13, 21, 34, 6, 6, 32, 5, 6]
```

Large-Input

```
testcase Total = 1
submarine.out: [197]
```